INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

NETWORK MEASURING SYSTEM INCLUDING A NETWORK ANALYZER, 500 KHz – 1.3 GHz WITH 50 OHM TEST PORT IMPEDANCE, CALIBRATION KITS, AND ACCESSORIES

The General Service Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE. This commercial item description (CID), describes a complete Network Measuring System which contains:

    a. Network Analyzer which operates over a frequency range of 500 KHz to 1.3 GHz with 50 Ω test port impedance.
    b. Fault isolation software.
    c. Calibration kits.
    d. Test cables and adaptors.

This CID is meant as a minimum requirement for the Network Measuring System in which only those manufacturers meet or surpass the following requirements are qualified per this CID.

2. CLASSIFICATION. The equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet the following description. The equipment shall be Class 4, in accordance with MIL-PRF-28800.

3. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and be bid sample tested within the accuracies, limits, and specifications herein.

3.1 Safety. The equipment shall comply with the safety requirements of MIL-PRF-28800 for the classification stated herein.

3.2 Environmental. The equipment shall meet all environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: The Engineering Division, 742 CBSSS/GBEC, 460 Richard Ray Blvd Suite 200, Robins AFB, GA 31098-1813. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

AMSC N/A FSC 6625

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3.3 Specialized Performance Characteristics. This specification will detail the requirements of this equipment in three main sections: System, Output Test Port, and Input Test Port.

3.3.1 System. The following specifications deal directly with the base system.

3.3.1.1 Frequency Range. The network analyzer shall have a frequency range of 500 kHz to 1.3 GHz with a minimum resolution of 1 Hz.

3.3.1.2 Measurement Capability. The equipment shall be capable of the following measurements:

- Gain/Directivity (dB)
- Insertion Loss (dB)
- Insertion Phase (degrees)
- Complex Transmission Coefficients (S12, S21) magnitude and phase (degrees)
- Separation of Transmission Components (Real and Imaginary)
- Electrical Length (m)
- Electrical Delay (s)
- Deviation from Linear Phase (degrees)
- Group Delay (s)
- Return Loss (dB)
- Complex Reflection Coefficients (S11, S22) magnitude and phase (degrees)
- Reflection Coefficients vs. Distance (Fourier Transform)
- Impedance (R + j X)
- SWR

3.3.1.3 System Performance. The equipment shall meet the following requirements when measured at the test ports after applying the error corrections, with no more than 20 averages, using OSLT with sliding loads, at 23 ± 5 degrees Celsius.

	700 KHz	**1 GHz**
Directivity (dB)	≥ 46	≥ 46
Source Match (dB)	≥ 40	≥ 40
Load Match (dB)	≥ 46	≥ 46
Reflection Tracking (dB)	±0.04	±0.04
Transmission Tracking (dB)	±0.06	±0.06
Dynamic Range (dB)	≥ 80	≥ 120

3.3.1.4 Measurement formats. Each channel shall be capable of measuring in the following formats: dB magnitude, linear magnitude, phase, polar, real, imag, Smith chart, group delay, SWR, inverted Smith chart, unwrapped phase

3.3.1.5. Data markers. Data Markers shall be easy to use and display in different formats. A minimum of eight markers, per tracing is required. The markers shall be searchable, as a minimum, by couples, max, min, peak and target.

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3.3.1.6 Trace Evaluation. The equipment shall be able to measure, as a minimum, the following: peak to peak, mean, and standard deviation up to ten definable ranges.

3.3.1.7 Measurement Speed and Data Transfer Times. The following parameters are required, as a minimum:

3.3.1.7.1 Measurement time. The measurement time shall be less than 6 ms in non-CW modes.

3.3.1.7.2 Measurement time per point. The measurement time in the CW mode shall be less than 4.5 µsec.

3.3.1.7.3 Data Transfer time. For all measurement points the data transfer time shall be as listed below:

Via IEC/IEEE bus	< 2.9 ms
Via VX11 over 100Mbits/s LAN	< 1.3 ms
Via RSIB over 100 Mbit/s LAN	< 0.7 ms

3.3.1.7.4 Switching time between Channels. The switching time between channels for the first 1500 points shall be less than 1 ms.

3.3.1.7.5 Switching Time between two preloaded instrument channels. The switching time between two preloaded instrument channels for the first 1500 points shall be less than 10 ms.

3.3.2 Test Port Output (Also known as the Source) Characteristics.

3.3.2.1 Test Port Output source. A built in synthesized source with a frequency range of at least 500 kHz to 1.3 GHz and 1 Hz resolution is required for the network analyzer. The expected overall source output power range of at least -40 dBm to +10 dBm. The level accuracy shall be at least +2 dB.

3.3.2.2 Test Output Port Power. The following table contains the power requirements, as a minimum, for this item:

	700 KHz	**1 GHz**
Power accuracy, referenced at -0 dB	< 2 dB	< 2 dB
Power Linearity, referenced at -0 dB	< 2 dB	< 2 dB
Power Resolution dB	0.05dB	0.05 dB
Harmonics (referenced to -5dB)	N/A	< -20 dBc

3.3.2.3 Measurement Accuracy. Specifications are based on a matched DUT, a measurement bandwidth of 10 Hz, and a nominal source power of -10 dBm. With the appropriate calibration kit the following is the accepted measurement accuracy for this item:

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Transmission Measurements		
300 kHz to 1 MHz	+15 dB to -45 dB	< 1 dB or < 6°
1 MHz to 50 MHz	+15 dB to -30 dB	< 0. 2 dB or < 2°
	-30 dB to -45 dB	< 1 dB or < 6°
above 50 MHz	+15 dB to +5 dB	< 0. 2 dB or < 2°
	+5 dB to -55 dB	< 0. 1 dB or < 1°
	-55 dB to -70 dB	< 0. 6 dB or < 4°
	-70 dB to -85 dB	< 2 dB or < 10°
Accuracy of Reflection Measurements		
above 300 MHz	+10 dB to +3 dB	< 0.6 dB or < 4°
	+3 dB to -15 dB	< 0.4 dB or < 3°
	-15 dB to -25 dB	< 1 dB or < 6°
	-25 dB to -35 dB	< 3 dB or < 20°
Trace stability (measured at 0 dB source power, 0 dB reflection, and 1 kHz measurement bandwidth)		
Trace Noise of S11 (rms)	Above 300 kHz	< 0.04 dB

3.3.3 Input Test Port.

3.3.3.1 Minimum specifications. The following table contains the minimum specifications of the Input Test Port.

	700 KHz	**1 GHz**
Source Match (dB)	≥ 16	≥ 16
Maximum Nominal Input Level (dBm)	+ 10	+ 10
Power Measurement Accuracy (dBm)	< 1	< 1
Receiver linearity (dB)	< 0.1	< 0.1
Damage Level (dBm)	+ 25	+ 25
Damage DC voltage (V)	30	30
Noise Level (dBm) (referenced to 10 Hz bandwidth). Noise level is defined as the rms value of the specified noise floor.	-70	-110

3.4 Test Ports. The instrument shall contain two test ports as a minimum. The test port impedance shall be 50 Ω. The connectors shall be type N connector and female.

3.5 Test Sets. The instrument shall allow forwards and reverse testing without reversing the test item.

3.6 Location of Connectors. The test port connectors shall be located on the front panel.

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3.7 S-Parameter Measurements. The equipment shall be capable of measuring all four two-port complex scattering parameters, also known as S-parameters.

3.8 Analyzer's Physical Aspects.

3.8.1 Dimensions. The equipment shall not exceed the maximum dimensions required for mounting in a standard 19.00 inch equipment rack.

3.8.2 Front Handles. This equipment shall contain front handles for bench top testing easy access.

3.9 Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard single-phase power sources with a voltage selectable from 110/120/220/240 Vrms AC ± 5% at line frequencies of 50 or 60 Hz ± 5%. Transient state conditions shall meet or exceed MIL-PRF-28800 requirements for the classification specified herein.

3.10 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 10,000 hours of operation with a statistical certainty of 98%. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion. Records of the actual equipment model shall be used, unless the Government authorizes the substitution of records from a similar model, due to the lack of sufficient supporting documentation on the offered equipment. The manufacturer shall be able to reasonably establish the claim that the offered equipment shall provide reliability similar to the model used to provide substitute records.

3.11 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the equipment case or modules.

3.12 Temperature Requirements.

3.12.1 Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 23º C to ±5º C.

3.12.2 Non-operating (storage) Temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment -8 º to +60 º C.

3.13 Warm Up Time. The equipment shall meet all of its specified parameters within a period of 90 minutes after it has been turned on at room temperature in a controlled environment.

3.14 Display. The following are required specification for the display:

(1) Full color LCD
(2) The equipment shall be capable of displaying a different S-parameter in each of the two display channels and a different graph type for each channel.
(3) The equipment shall be capable of displaying polar, rectilinear, or Smith Chart graphs for the frequency domain.

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(4) The rectilinear graph types shall include, at a minimum: magnitude, phase, SWR, group delay, real, and imaginary. The equipment shall be able to display magnitude in linear or logarithmic format.
(5) The equipment shall be able to display time domain response as a function of time or distance.

3.15 I/O Ports**.** The instrument shall have as a minimum one Ethernet and two USB Input/Output Ports.

3.16 Software/Firmware Requirements. The equipment shall include the following software features.

3.16.1 Adapter Removal. The equipment shall include calibration software or firmware to compensate for the electrical effects of adapters on measurements.

3.16.2 LRL, LRM, or TRL Calibration. The equipment shall include software or firmware to support full LRL/LRM or TRL/TRM calibrations.

3.16.3 Fault Isolation Software. The equipment shall include software or firmware to support fault isolation analysis. The analysis will provide distance and time domain information including bandpass, lowpass impulse, and lowpass step measurements/calculations.

3.17 Calibration Types and Accuracy.

3.17.1 Calibration Types. The equipment with the appropriate accessories shall have the capability for the following types of calibration:

- OSLT - Open-Short-Load-Through using short circuits, open circuits, terminations (fixed and/or sliding)
- LRL/LRM or TRL/LRM - Line-reflect-line/line-reflect-match or thru-reflect-line/line-reflect-match
- Offset short (waveguide)

3.17.2 Error Correction Models. The equipment with the appropriate accessories shall have the capability to collect data for and to use the following error correction models:

- full two-path/two port error correction
- one path/two port error correction
- frequency response (transmission/reflection) error correction
- reflection only error correction

3.18 Calibration Kits. The following calibration kits (mechanical or electronic), OSLT, sliding loads optional, and gauges, shall be included for the RF range 500 kHz to 1.3 GHz, 2 ports:
(1) Type-N (f) to Type-N (m)
(2) Type-N (m) to Type-N (m)
(3) Type-N (f) to Type-N (f)

These calibration kits may be used to calibrate components with the specific connectors listed. Any proposed substitutions shall be mechanically compatible with the kits specified herein.

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Third party calibration kits are acceptable as long as they are included in this submission's parts list.

3.19 Test port cables. This system shall include the following test port cables to be used in conjunction with testing the unit under test and the calibration kits for a complete measurement system:

(1) Sturdy (not economy) 50 Ω Type-N cable between 22 and 30 inches in length with male connectors.
(2) Sturdy (not economy) 50 Ω Type-N cable between 22 and 30 inches in length with both male and female connectors.
(3) Sturdy (not economy) 50 Ω Type-N cable between 48 and 60 inches in length with male connectors.
(4) Sturdy (not economy) 50 Ω Type-N cable between 48 and 60 inches in length with both male and female connectors.

3.20 Test Cable Adapters. The system shall include the following adapters:

(1) Sturdy (not economy) 3.5 mm (f) to 50 Ω Type-N (m)
(2) Sturdy (not economy) Type-N (f) to Type-N (f)
(3) Sturdy (not economy) Type-N (m) to Type-N (m)
(4) Sturdy (not economy) Type-N (f) to 7 mm
(5) Sturdy (not economy) Type-N (m) to 7 mm

3.21 Specialized Tools. The system shall include any tool kits generally used on this system. These items may constitute the following and shall be listed in the parts listing.

(1) specific torque wrenches,
(2) centering bead,
(3) gage sets,
(4) offset opens and shorts, broadband loads and sliding load terminations,
(5) service and user guide,
(6) calibration kit storage case

3.22 Verification Kits. The following calibration verification kits shall be included for the RF range 500 kHz to 1.3 GHz, 2 ports. Third party Verification kits are acceptable as long as they are included in this submission's parts list.

3.23 External reference signal input connector. The External reference signal input connector shall be a BNC type, female. The input frequency shall be 10 MHz ± 10 ppm with an input power level of 0 dBm ± 3 dB.

3.24 Connector Type. The equipment shall have a input test port and a output test port capable of operating within the frequency range and other capabilities specified for the equipment herein. Input and output ports shall use the same type of connector.

3.25 Access to non-ratioed channels**.** This item will provide direct access to all non-ratioed channels from the front panel of the Vector Network Analyzer.

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3.26 Manuals. The synthesizer shall be delivered with operator, maintenance, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order. Contractor format is acceptable.

4. REGULATORY REQUIREMENTS.

4.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE

5.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 The contracting officer has the option of accepting or rejecting the product.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Sources of documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://assist.daps.dla.mil or from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 The Federal Acquisition Regulation (FAR) may be obtained at www.acqnet.gov/far or from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

7.2 Ordering data. The contract or order should specify the following:

a. CID number with revision date.
b. Product conformance provisions.
c. Packaging requirements.

7.3 Key words.

a. digital input/output
b. output connector
c. front handle kits
e. isolation software

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Preparing activity:
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